INTRODUCTION

Metallurgy relate to the science and technology of metals. In nature, only a few metals occur in their native state, all other metals occur in a combined state as their oxides, sulphides, silicates etc... The extraction of pure metals from their natural sources, is linked to the history of human civilisation. Ancient people used the available materials in their environment which includes fire and metals, and they were limited to the metals available on the earth's surface. In the modern world, we use a wide range of metals in our daily life, which is the result of the development of metallurgical knowledge over thousands of years. Our need for the materials with specific properties have led to production of many metal alloys. It is essential to design an eco-friendly metallurgical process that would minimize waste, maximize energy efficiency. Such advances in metallurgy is vital for the economic and technical progress in the current era. In this unit we will study the various steps involved in the extraction of metals and the chemical principles behind these processes.

1.1 Occurrence of metals

In general, pure metals are shiny and malleable, however, most of them are found in nature as compounds with different properties. Metals having least chemical reactivity such as copper, silver, gold and platinum occur in significant amounts as native elements. Reactive metals such as alkali metals usually occurs in their combined state and are extracted using suitable metallurgical process.

1.1.1 Mineral and ore

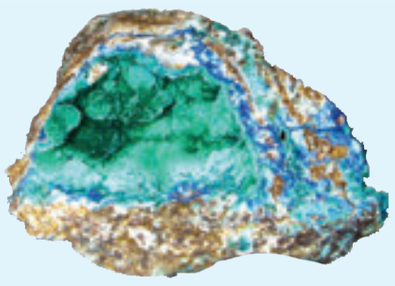
A naturally occurring substance obtained by mining which contains the metal in free state or in the form of compounds like oxides, sulphides etc... is called a **mineral**. In most of the minerals, the metal of interest is present only in small amounts and some of them contains a reasonable percentage of metal. For example iron is present in around 800 minerals. However, some of them such as hematite magnetite etc., containing high percentage of iron are commonly used for the extraction of iron. Such minerals that contains a high percentage of metal, from which it can be extracted conveniently and economically are called **ores**. Hence all ores are minerals but all minerals are not ores. Let us consider another example, bauxite and china clay ($Al_2O_3.SIO_2.2H_2O$). Both are minerals of aluminium. However, aluminium can be commercially extracted from bauxite while extraction from china clay is not a profitable one. Hence the mineral, bauxite is an ore of aluminium while china clay is not.

The extraction of a metal of interest from its ore consists of the following metallurgical processes.

(i) concentration of the ore

(ii) extraction of crude metal

(iii) refining of crude metal



Malachite – copper mineral